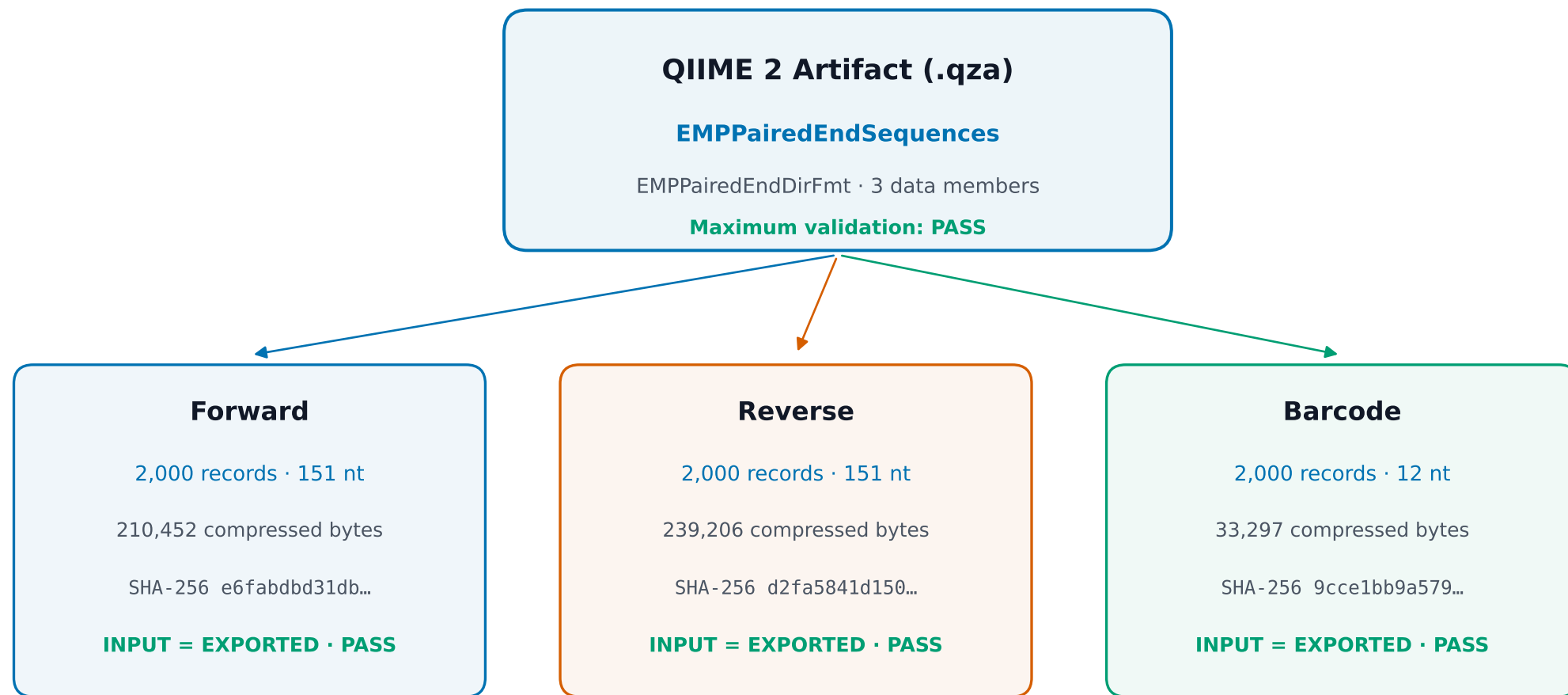


Real Atacama Artifact round-trip audit

Import → maximum validation → export; compressed FASTQ bytes remain unchanged.



The .qza UUID and archive hash change on a new import; the semantic type, format, provenance, and exported-data hashes are the stable contract.